Weimin Yuan

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Personal Home Page
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Research Area: Deep Learning, Game Theory, LLM, AI Application

Education Experience

•**Bachelor of East China Jiaotong University (ECJTU)** *2017-2021*
Major in Computer Science and Technology **GPA:** 3.78/5.0; **Ranking:** 2/66

•**Master from University of Electronic Science and Technology of China (UESTC)** *2021-2024*
Major in Computer Technology **GPA:** 3.84/4.0; **Ranking:** 32/465

•**PhD candidate at the University of Houston (UH)** *2024-Present*
Major in Electrical Engineering **Advisor: Dr. Zhu Han**

Selected Publications

Journal:

- Han Yang, **Weimin Yuan***, Weijun Zhu, Zhenye Sun, Yanru Zhang, and Yingjie Zhou, "Wind turbine airfoil noise prediction using dedicated airfoil database and deep learning technology," *Applied Energy*, 2024.
- Weilong Chen, Wenhao Hu, Xiaolu Chen, **Weimin Yuan**, Yan Wang, Yanru Zhang, and Zhu Han, "Tri-modal transformers with mixture-of-modality-experts for social media prediction," *IEEE Transactions on Circuits and Systems for Video Technology*, 2024.
- Yuqi Liu, **Weimin Yuan**, Weilong Chen, Wenming Li, Han Yang, and Yanru Zhang, "CPLLM-WPF: A multi-scale prompting framework for generalizable wind power forecasting with LLMs," *Applied Energy*, 2025.
- **Weimin Yuan**, Han Yang, Yanru Zhang, and Zhu Han, "Enhancing wind power forecasting accuracy under extreme weather: Leveraging a dual-model approach with condition-based classification," *Engineering Applications of Artificial Intelligence*, 2025.
- Alexander Tang, **Weimin Yuan**, Shiwen Mao, Dusit Niyato, and Zhu Han, "Spatio-Temporal Latent Diffusion Inpainting for UAV Radio Map Reconstructions," *IEEE Transactions on Vehicular Technology*, 2025.

Conference:

- Weilong Chen, Chenghao Huang, **Weimin Yuan**, Xiaolu Chen, Wenhao Hu, Xinran Zhang, and Yanru Zhang, "Title-and-Tag Contrastive Vision-and-Language Transformer for Social Media Popularity Prediction," *ACM International Conference on Multimedia (ACM MM)*, 2022.
- Chenghao Huang, Xiaolu Chen, Yuxi Chen, Yutong Wu, **Weimin Yuan**, Yan Wang, and Yanru Zhang, "A generic pre-trained BERT-based framework for social media health text classification," *Proceedings of the Seventh Workshop on Social Media Mining for Health Applications, Workshop & Shared Task*, 2022.
- Yuxi Chen, Chenghao Huang, **Weimin Yuan**, Yutong Wu, Ziyi Zhang, Hao Tang, and Yanru Zhang, "Inference-Based Deep Reinforcement Learning for Physics-Based Character Control," *IEEE DependSys*, 2022. (Best Paper Award)
- Wenhao Hu, Weilong Chen, **Weimin Yuan**, Yan Wang, Shimin Cai, and Yanru Zhang, "Dual-stream pre-training transformer to enhance multimodal learning for social media prediction," *ACM International Conference on Multimedia (ACM MM)*, 2024.
- Wenhao Hu, Weilong Chen, **Weimin Yuan**, Xiaolu Chen, Han Yang, Yanru Zhang, and Zhu Han, "Feature Disentangling Dual-stream Network for User Bias Alleviation in Social Media Prediction," *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2025.
- **Weimin Yuan**, Weilong Chen, Hien Nguyen, Yifei Zhu, Dan Wang, and Zhu Han, "Integrating Mean-Field Game Theory with Diffusion Model," 2025 IEEE 26th International Workshop on Signal Processing and Artificial Intelligence for Wireless Communications (SPAWC), 2025.

Under Review:

- **Weimin Yuan**, Dingwen Pan, Hien Van Nguyen, Yifei Zhu, Dan Wang, Han Yang, and Zhu Han, "Deception in Large Language Models: An Audit Game-Theoretic Analysis," under review, *IEEE Transactions on Network Science and Engineering*, 2025.
- **Weimin Yuan**, Dingwen Pan, Hien Van Nguyen, Yifei Zhu, Dan Wang, Han Yang, and Zhu Han, "Mean-Field Reinforcement Learning for Large-Scale Populations," under review, *NIPS*, 2026.
- Dingwen Pan, **Weimin Yuan***, Chenye Wu, Dan Wang, Hien Van Nguyen, and Zhu Han, "Learning Collusion-Resistant Auctions via Coalition Regret Minimization," under review, *NIPS*, 2026.
- **Weimin Yuan**, Lu Gao, and Zhu Han, "ROAD-Former: A Robust Observation-Attribute Decoupling Transformer for Network-Level Pavement Deterioration Forecasting," under review, *Transportation Research Part C*, 2026.

Research and Project Experience

•Machine Learning for Spatial Dynamic Wind Power Forecasting. *Apr-Jul 2022*

- Visualize, analyze, and preprocess data. Considering turbine location and data features, apply clustering methods and build tree models for time series analysis.
- Explore the implementation of Informer, LSTM and other models to achieve higher accuracy prediction.
- The ranking on KDD CUP 2022 is **40/2490** (Team leader).

•SMP Challenge @ ACM Multimedia. *2022-2024*

- Participated in the 4th–6th SMP Challenge, focusing on large-scale social multimodal prediction tasks.
- Led competition modeling and feature analysis for multimedia, text, structured user information, and social media popularity signals.
- Achieved Top Performance (Second Prize) awards in all three consecutive competitions.

•Parkinson's Disease Progression Prediction @ Kaggle. *Feb-May 2023*

- Predicted longitudinal MDS-UPDRS scores from protein and peptide-level data using feature engineering, neural networks, and LightGBM models.
- Utilized Powell method for searching optimal trend terms and adopted quadratic regression, model fusion, and cross-validation for robust prediction.
- Achieved **Rank 37/1805** and received a Kaggle Silver Medal.

•Power Grid Load Forecasting. (Project of State Grid Hunan Electric Power Company) *Oct-Dec 2023*

- The project is overall divided into three modules: identifying abnormal data module, correcting abnormal data module, and analyzing and completing empty data module ((Project leader)).
- Detecting outliers through box plots and the K-means algorithm. Filling missing data using interpolation. Applying RNN, LSTM, and GRU to correct abnormal data. Design and build the UI interface for program execution.

Honors and Awards

•First-class Scholarship of ECJTU & UESTC *2019, 2020, 2021, 2022, 2023*

•Top Performance Award of ACM Multimedia Social Media Prediction Challenge *2022, 2023, 2024*

•IEEE DependSys Best Paper Award *2022*

•Kaggle AMP Competition is 37/1805 (Silver Medal) (Individual participant) *2023*

• The 7th China College Students' "Internet+" Innovation and Entrepreneurship Competition (Silver Medal) *2022*

•College Computer Skills Challenge Second Prize in the East China Region (C++) *2020*

Technical Skills and Interests

- **Coding Languages**: Python, C/C++/C#, Java; HTML+CSS.
- **Operating Systems**: Linux; - **Database**: SQL.
- **Deep Learning Frameworks**: PyTorch, TensorFlow.
- **LLM Framworks**: GPT-3.0, Llama, DeepSeek-V3.
- **Areas of Interest**: Machine Learning, LLM, MultiModal, Game Theory, Data Mining and Analysis.